V3: Reasoning as a Path of Least Action

An Inference-Time Thought-Anchor Detector
Built on the Momentum–Work Analogy

Working specification — conceptual base, validity tests, branch-conditional models, and implementation plan

Abstract

The *task* of this work is **online detection of thought anchors**: predicting, *during* chain-of-thought generation and with only bounded lag, which reasoning steps will turn out to have high *counterfactual influence on the final answer* in the sense of Bogdan et al. (arXiv:2506.19143). All existing anchor-attribution methods are retrodictive — they require the full trace or many rollouts — so to our knowledge no causal, generation-time predictor of counterfactual anchor-influence exists by *any* method. That empty cell, not the physics, is the primary contribution claimed here. The *method*, "V3," is one instantiation: treat the trace as the trajectory of a mechanical system and flag steps where momentum is sharply reinforced or redirected. V3 is built around an honest commitment — the mechanical analogy is admissible only if the dynamics support it, so the method is gated by three falsifiability tests and the downstream model is *selected by the test outcomes* rather than assumed. We separate the task from the method throughout, place both against the recent trajectory-geometry literature (§3), and organize the whole program as a **tiered execution plan** (§7): a minimal first run carrying only the non-negotiable safeguards, then small, outcome-driven increments — fixes if a tier fails, improvements if it succeeds — until either every necessary fix has been tried and failed, or the best working version is reached. We state plainly that the load-bearing requirement is validation against *counterfactual-importance labels* (not sentence-type, correctness, or difficulty), and give the conceptual base, the exact tests, the branch-conditional model for each outcome, and variable-level implementation details. Red boxes throughout give a strict account of what is principled, what is a modeling choice, what is numerically fragile, and what no implementation can rescue.

Honesty preamble. Nothing here has been validated on a real model. The synthetic-data sanity checks confirm only that the arithmetic does what the equations say, not that the equations describe a real reasoning model. Two axes should be kept separate. On the *dynamics* axis, the most likely landing is the dissipative case (§5, Branch B), not the clean conservative one. On the *outcome* axis — does any version beat the baselines — the most likely result is still a *negative*: even the right branch may fail to track anchors above what trivial features capture (see §9). The two are compatible: "probably dissipative, and probably still does not beat baselines." That negative is a result, not a failure, and the value of the design is that it reaches the verdict cheaply. Read every red box; the boxes are where the real uncertainty lives.

Contents

1 Conceptual base

1.1 What the detector is trying to predict

A thought anchor is defined *retrodictively*: a reasoning step is an anchor because of its downstream effect on the final-answer distribution, as measured by counterfactual resampling. At inference time that downstream effect has not happened yet, so anchor-ness is not directly observable online. The entire premise of this work is therefore a *bet*: that anchors carry a *local dynamical signature* in the model's representation trajectory that is predictive of the retrodictive label. We look for a causal (past-and-present, possibly lagged) functional of the trajectory that correlates with the future-defined anchor.

This bet can simply be false. If anchor-ness is genuinely determined by what the model does *after* a step — not encoded in the local trajectory shape — then no causal functional can predict it above a low ceiling, and the ceiling is estimable (§9). We design to maximize detection *if* the signal exists; we cannot manufacture it if it does not.

1.2 The mechanical analogy

We model the trace as a particle moving through representation space. Let $h_t \in \mathbb{R}^d$ be the model's hidden state (residual stream) at step t, and $v_t = h_t - h_{t-1}$ the velocity. We endow the space with a metric G (the mass tensor), a potential $V(h)$, and the Lagrangian

$$L \;=\; \tfrac{1}{2}\, v^\top G\, v \;-\; V(h), \qquad p \;=\; \frac{\partial L}{\partial v} \;=\; G\, v. \tag{1}$$

This is a *natural mechanical system* in the sense of Arnold: momentum is the covector $p = Gv$, force is $-\nabla V$, and (for a time-independent L) the energy $E = \frac{1}{2}v^\top G v + V$ is conserved along true trajectories.

1.3 The defining commitment of V3: the potential

What makes this *V3* rather than a generic momentum heuristic ("V2") is the potential V. We fix

$$V(h) \;=\; -\log P_\theta(\text{realized next step} \mid h), \tag{2}$$

the surprisal of the continuation that was actually generated, evaluated as a function of an arbitrary spliced-in state h. Low potential means the model is rolling toward high-probability continuations; high potential means it is paying probability cost. Coasting deduction is then motion under inertia plus this potential; an anchor is a step the potential-plus-inertia dynamics cannot explain — a non-conservative injection.

Equation (2) ties V to the *realized* path, so it is a potential defined relative to the trajectory taken. A genuine physical potential is a field that exists independently of the particle. This is exactly why the curl test (§4) is non-negotiable: a path-relative "potential" has no guarantee of being curl-free, i.e. of being the gradient of any scalar at all. Surprisal is also only *one* defensible potential; a value- or reward-model potential would be different and possibly better, and is chosen here only because it is available at inference without a second model.

1.4 The load-bearing reframe: impulse, not work

The momentum–work analogy has a robust half and a fragile half, and the distinction organizes the entire branch tree below.

- **Impulse** $\boldsymbol{J} = \int \boldsymbol{F}\, dt = \Delta p$ is a vector: the change in momentum. It is the robust primitive.
- **Work** $W = \int \boldsymbol{F} \cdot \boldsymbol{v}\, dt$ is a scalar: it is impulse *projected onto the direction of motion*, $W = \boldsymbol{J} \cdot \hat{v}$. It discards any force perpendicular to v.

Consequently a force that bends the trajectory without speeding it up has full impulse and *zero* work. Every branch where the "momentum–work analogy" breaks is a branch where the anchor force is partly perpendicular to v, and work goes blind to exactly that component. **The detector is therefore built on the impulse residual; work is reported only as a derived, lossy scalar where it is meaningful.**

1.5 Task versus method, and the non-negotiable label

Two things must be kept apart for the rest of this document to be read correctly. The **task** is online prediction of *counterfactual answer-influence*: flagging, during generation, which steps would — if resampled — most change the final-answer distribution. The **method** (V3, the mechanical model) is one way to attempt the task; it is not the task, and it can fail while the task remains well-posed and approachable by other means. Consequently the only valid ground truth is the resampling-derived counterfactual-importance score of Bogdan et al. Validating against sentence *type* (planning/backtracking), *correctness*, or *difficulty* would silently substitute an easier, different target; because those proxies are already known to be decodable from trajectory geometry (§3), matching them would prove nothing about anchors. Every accuracy claim in this document is therefore implicitly "versus counterfactual-importance labels on the same traces."

> This is the easiest place in the whole program to fool oneself. "Planning sentences are anchors" is a finding of the original paper, and planning sentences are easy to detect; a detector that fires on them will look successful while having learned nothing about counterfactual influence. If labels are not the resampling scores, the result is not about anchors.

1.6 The anchor unit: clause-respecting token windows

A point of departure from Bogdan et al.: although we adopt their counterfactual notion of importance, the unit of analysis is *not* the sentence. An anchor here is a token-window span $\mathcal{S}_i$ — but the windows are not cut at an arbitrary token count. The motivation is twofold: small reasoning models emit short traces, so sentence units give too few labels per trace and coarse localization; yet a window that ignores syntax can split a single fact, relation, or asserted value across two spans and dilute its measured importance. We therefore use a **two-stage** definition that respects linguistic structure before windowing.

Stage 1 — parse into clauses. Segment the trace into sentences, then into clauses, splitting at sentence terminators, newlines, comma-delimited clause boundaries, and discourse connectives (*so, therefore, thus, then, because, . . .*). Each resulting clause is a self-contained linguistic unit.

Stage 2 — window within clauses. Subdivide each clause into contiguous windows of at most τ tokens. Windows never cross a clause boundary: a clause shorter than τ is a single span; a longer clause becomes $\lceil \mathrm{len}/\tau \rceil$ spans, with a short trailing fragment merged into its neighbor within the same clause. The window size τ is swept once and then held fixed.

This makes a span always a sub-unit of one clause, never a fragment straddling two. Throughout the implementation (§6), wherever the physics refers to a "step" or a "boundary" it means one such clause-respecting window and its right edge, not a sentence (note the time/step index t in the physics is unrelated to the window size τ used here); the physics is unchanged because all per-token quantities aggregate over arbitrary spans.

Two knobs now exist where there was one: the clause-parse rule (Stage 1) and the window size τ (Stage 2), and both can change the labels. Respecting clause boundaries removes the worst failure of arbitrary windowing (a fact split across two spans), but it does not eliminate the dependence — a clause longer than τ is still cut internally, and the clause parser itself can mis-split (nested clauses, lists, math). Fix both rules once, *before* any signal is fit, and report them; sweeping span geometry alongside the detector would let it launder signal. The constructive sanity check (§2.5) is the guard: if a known supporting fact lands cleanly inside one span under the chosen τ, the parse is doing its job.

2 Dataset generation: token-window anchor labels

The detector needs ground-truth anchor labels to validate against. This section adapts a parallel protocol's data-generation pipeline (perturbed parallel resampling at word-group granularity) to the token-window setting, and adds two elements specific to the present method: storing full traces with token alignment, and recording a *first-content-token* probe decision per span. The pipeline is independent of the physics — it produces the labels any online anchor detector (physics or learned) would be scored against.

2.1 Perturbed resampling labels

For span i in a base trace, importance is the counterfactual dependence of the answer on i's content. Generate R continuations from the prefix *through* i ("real") and R from a prefix in which i is replaced by a perturbation i', and measure the shift in the final-answer distribution. The perturbation must be *genuinely different in meaning* from i, or the comparison is not a counterfactual at all — so the primary procedure is the Thought-Anchors resample-and-filter: resample i' from the model, then keep it only if it passes a semantic-difference test (§2.2). A construction-time alternative is available on the synthetic curriculum, where the vocabulary is controlled: swap the entity, relation, or value asserted in i for another from a mutually exclusive set ("the box is in the kitchen" $\to$ "... in the garden"), which guarantees a different meaning by construction and needs no filter. We use the resample-and-filter as primary everywhere (it is the established method and applies to any task), and the controlled swap as a cheaper, filter-free cross-check on the synthetic tasks where it is available. For verifiable-answer tasks the answer is discrete, so the label is the total-variation distance between the real and perturbed answer distributions:

$$A_i \;=\; \mathrm{TV}\Big(\hat{p}_{\text{ans}}\,|\,\text{real}_i,\;\; \hat{p}_{\text{ans}}\,|\,\text{perturbed}_i\Big) \;=\; \tfrac{1}{2}\sum_a \big|\hat{p}(a \mid i) - \hat{p}(a \mid i')\big|, \tag{3}$$

with large A_i marking an anchor. Pipeline: (1) generate base traces; (2) parse into clauses, then window within clauses (§1.6); (3) per span, dispatch $2R$ rollouts (R real, R perturbed), batched across GPUs; (4) aggregate answer distributions and compute A_i; (5) in parallel, hook the forward-pass quantities the detector needs (hidden states for h_t, output distributions for V and the KL proxy). The independent unit for statistics is the *trace*, not the span, since within-trace spans are correlated.

This is the gold label and it is expensive: $2R$ rollouts per span $\times$ spans per trace $\times$ traces. It is also the *only* non-circular ground truth, because it is computed from the text (black-box) and is independent of the forward-pass signals being validated. A cheaper hidden-state proxy (e.g. a finite-time trajectory-divergence / Lyapunov score) may be calibrated against A_i and used to scale labeling *only after* a high measured correlation — and it carries a circularity risk precisely because it reads the same internal states the detector does. Resampling stays primary.

2.2 Semantic-difference adjudication: which resamples count

A resampled i' only yields a valid counterfactual if it *means something different* from i; a paraphrase contributes a spurious zero-importance reading and corrupts A_i. The procedure that decides "different vs. not" is therefore load-bearing, and we adopt a tiered set of options trading compute for fidelity.

The primary *procedure* is the Thought-Anchors resample-and-filter; within it, the adjudicator that decides "different vs. not" is one of the following, ordered by compute. The *default* adjudicator is the cosine-median filter; NLI is an *upgrade* used where the budget is ample, and the rest are fallbacks for when it is not.

Default adjudicator — Sentence-BERT median-threshold filter (the Thought-Anchors method). Embed i and each resample i' with Sentence-BERT; keep i' as a valid "different" counterfactual only if $\cos(i, i')$ falls *below the median* cosine similarity over all span pairs in the dataset (the threshold is a single precomputed dataset-wide scalar). This is the established definition of semantic dissimilarity from the original method, and — importantly for the compute story — it is *cheap*: one embedding pass per resample plus a precomputed median. The expensive part of labeling is the R resampling rollouts, which are paid regardless of how difference is adjudicated; the filter itself adds negligible cost. Resamples that fail the filter are discarded and re-drawn up to a fixed attempt budget, so each span accumulates R *filtered* perturbed rollouts. This is the adjudicator carried into the primary experiment and the only one that scales unchanged to large models.

Upgrade — NLI non-entailment (where compute is ample, e.g. the small target model). "Below-median cosine" is a blunt proxy: it can pass a surface edit that preserves meaning or reject a genuine change phrased similarly. Where the budget allows — chiefly on the small target model (DeepSeek-R1-Distill-Qwen-1.5B scale), where a second model is cheap to hold — upgrade the adjudicator to require that i' be *not entailed* by i (and ideally mutually non-entailing) under a natural-language-inference model: a sharper test of "different meaning" than embedding distance, at the cost of one NLI forward pass per candidate. This is a strictly higher-fidelity replacement for the cosine filter, not a different procedure; it is optional, and if used it is fixed for the whole label set (see the hazard below). **The headline experiment uses the cosine-median filter as its adjudicator** — it is the established method, it scales unchanged, and it keeps the label definition identical across model sizes; NLI is run, if at all, only as a fidelity cross-check on the small target model, never mixed into the primary label set.

Lower-compute fallbacks (for larger models, when cost gets out of hand). Even with the cheap cosine filter as adjudicator, two things bite at scale: the re-draw loop (rejected resamples waste rollouts) and, *if* the NLI cross-check was being run, a second model held in memory beside a large target. Fall back, in order:

1. **Drop the NLI cross-check, keep the cosine median filter.** Removes the second model entirely; the embedding pass is negligible. The headline adjudicator is unaffected — this only retires the optional fidelity check.
2. **Construction-time difference on the synthetic curriculum.** Where the vocabulary is controlled, draw i' from a mutually exclusive set so difference is guaranteed and *no filter and no re-draw* are needed — the cheapest path, available precisely on the tasks used for the early constructive check (§2.5).
3. **Cache and reuse the embedding model** across all spans/traces (a small fixed model, e.g. a MiniLM-class encoder), and cap the re-draw budget so a span that cannot produce a passing resample is dropped rather than retried indefinitely.
4. **Cheaper encoder / precomputed threshold.** Swap Sentence-BERT for a smaller sentence encoder and freeze the median threshold from a calibration subset rather than recomputing over all pairs.

The filter choice changes the labels, so it must be fixed and reported, not tuned. Two specific hazards. First, the median threshold is dataset-relative: computed over *clause-respecting spans* (§1.6), not sentences, its value differs from the original paper's, so the threshold must be recomputed on our own span distribution, never borrowed. Second, swapping adjudicators between the primary run and a fallback is not free — cosine-median and NLI-non-entailment do not select the same resamples, so a label set built under one is not interchangeable with the other; if compute forces a switch mid-program, the affected traces must be re-labeled under the new rule, not mixed. Pick one adjudicator per label set and state which.

2.3 Trace storage with token alignment

Because the unit is a token window and the physics reads per-token hidden states, the stored artifact must preserve exact token alignment, not just text. For each base trace we store: the token id sequence (prompt and generation separated); the character/token offsets so spans map back to text; the per-token clause-boundary and window-boundary flags from the two-stage segmenter; the per-token surprisal (for V); the sampled generation-time hidden states at the chosen layer(s) and a stride; and, per span, the rollout answer distributions and the computed A_i. Storing token alignment is what lets the same archive serve resampling labels, the physics signals, and the baselines on a common index (item, trace, run, layer, token), so a span's label and its dynamical signals are never misaligned by a re-tokenization.

Token alignment is a frequent silent bug source: re-tokenizing stored text can yield a different token count than generation produced (whitespace, special tokens, BPE merges at span edges), which shifts every per-token signal relative to its label by a few positions — enough to destroy a real correlation or manufacture a spurious one. Store the *generated token ids directly* and never re-tokenize for alignment. This is mundane and load-bearing.

2.4 The first-content-token probe decision

The online constraint forces a decision about *when*, within a span, the detector commits. We record a **first-content-token** decision: the detector scores the span as soon as its first content-bearing token has been generated (skipping leading whitespace, punctuation, and pure discourse connectives), rather than waiting for the span to complete. This is the most stringent online setting — it maximizes lead time — and it is recorded as an explicit per-span decision point so that the achievable accuracy can be reported *as a function of* how early the commit is made. Concretely, for each span we store the token index of its first content token and evaluate the detector's causal signal up to that index (plus the fixed Savitzky–Golay lag of §6), giving an early, fixed-lead-time decision; the span-complete decision is retained as the lower-stringency comparison.

> Two honest points. First, the first-content-token decision is where the *causal ceiling* bites hardest: the less of the span the detector has seen, the less local-trajectory evidence exists, so early commit trades accuracy for lead time, and the trade curve — not a single number — is the honest result. Second, "content token" is a definitional choice (which tokens count as discourse-only) that can shift the decision index; fix the rule with the span definition (§1.6) and report it, because moving it changes both the lead time and the score.

2.5 Curriculum and the constructive sanity check

Labels are generated over a difficulty ramp of *verifiable, state-tracking* reasoning, the regime where anchors are most pronounced and pivotal steps are sequential and identifiable: synthetic state-tracking with known on-path supporting facts (a *constructive* ground truth independent of resampling — the load-bearing facts are labeled by construction, so signals and A_i should both rank them above distractors); then progressively harder state-tracking tasks (object-tracking under swaps, compositional kinship with graded hop count, long-horizon instruction execution). A *different* reasoning type (e.g. grade- school math) is held out purely as an out-of-distribution generalization test, not for fitting, so a positive result is not silently confined to state-tracking.

> The constructive check (synthetic tasks with annotated supporting facts) is the cheapest and most valuable early signal: it validates the whole pipeline — segmentation, alignment, label computation — against a ground truth that needs no resampling. If the resampling label A_i does not rank annotated supporting facts above distractors on these tasks, the pipeline is broken and no downstream result is trustworthy. Run it first.

3 Related work and the precise novelty claim

This work sits next to a fast-moving 2025–2026 literature on hidden-state *trajectory geometry*. The claim here is narrow and must be stated against that literature exactly, because the substrate ($v_t = \Delta h_t$) is shared while the *target* and the *causal/online constraint* are not.

The anchor target is retrodictive in all prior anchor work. Bogdan et al. (2506.19143) define anchors and give three attribution methods — counterfactual resampling ($\sim$100 rollouts), attention aggregation, and attention suppression — all of which require the full trace or many rollouts. None is causal/online. The task of *predicting* anchor-influence during generation is not addressed there or, to our knowledge, anywhere.

Substrate prior art (shared method, different target). A cluster of papers analyzes generation- or layer-time trajectories but targets something other than counterfactual answer-influence: Zhou et al. (2510.09782, ICLR 2026) model reasoning as smooth flows and use position/velocity/(Menger) curvature to study logic-vs-semantics — pure kinematics, no mass/momentum/energy/potential, and no detector; Damirchi et al. (2603.01326) learn a classifier on *layer-wise* displacements to detect reasoning *validity*; "TRACED" (2603.10384) decomposes trajectories into progress (displacement) and stability (curvature) for reasoning-*quality* assessment; Gjølbye et al. (2605.15454) study difficulty-vs-geometry coupling and establish that raw geometry is dominated by generation length. These share the kinematic substrate but not the anchor target; they are prior art on *trajectories*, not competitors on *online anchor isolation*.

Closest near-misses, and why they are not the task. Two works are genuinely online/predictive, which makes them the right *baselines* but not task-competitors. Sun et al. (2604.05655, ACL 2026) predict final-answer *correctness* mid-reasoning (ROC-AUC up to 0.87) via step-specific subspaces and add trajectory steering — online and predictive, but the target is correctness, not anchor-influence, and the mechanism is learned subspace geometry, not physics. Marín (2410.04415) is the one prior work that reaches a genuine *mechanical* formalism (a reasoning Hamiltonian $H = T - V$ with $p = \Delta q$), but it is *offline*, uses a goal-similarity potential that needs the answer (so it cannot run causally), predates Thought Anchors, and targets valid-vs-invalid reasoning. It also states plainly that the physics–reasoning connection "remains largely metaphorical" — the same caveat carried throughout this document.

The novelty claim, stated narrowly. Crossing target × causality × formalism: *online prediction of counterfactual anchor-influence* is unoccupied by *any* method; this is the primary claim. The mechanical/least-action *instantiation* of it is a secondary, separable claim. The shared kinematic substrate (Zhou, Damirchi, Gjølbye), online correctness prediction (Sun), and the offline mechanical framing (Marín) are all claimed elsewhere.

> The walls around the empty cell are close. Two consequences for honesty: (i) the contribution is *not* "trajectories/velocity for reasoning" — that is crowded, with work at ICLR and ACL — so the framing must lead with the *task*, not the geometry; (ii) the real baselines-to-beat are concrete and published: a learned step-boundary classifier trained against counterfactual-importance labels (the atheoretical version of the task), and a length-corrected kinematic signal in the spirit of Gjølbye et al. If the physics merely matches these, it is an elegant reframing, not a new capability. The collapse-to-V2 and collapse-to-kinematics ablations (§5, §9) exist precisely to detect that outcome. Usefully, the Gjølbye et al. trajectory archive and Sun et al.'s released code remove two of the three things one would otherwise have to build (an extraction substrate and a strong online baseline).

4 The three validity tests

The mechanical model is admissible only if the dynamics support it. Three tests, run *before* tuning any detector, decide which model (if any) is valid. They probe independent structure and can fail in compatible combinations.

Test 1 — conservativeness (coasting energy flatness). Compute $E_t = \frac{1}{2} v_t^\top G v_t + V(h_t)$ and its rate dE/dt over steps labeled as coasting (§6.6). *Pass:* $dE/dt \approx 0$ on coasting; the system admits a time-independent Lagrangian. *Structured fail:* dE/dt is nonzero but fittably patterned (e.g. steady decay)→ dissipative. *Unstructured fail:* dE/dt large and patternless → no energy concept survives.

Test 2 — conservativeness of the force (curl). Estimate the force field $F(h) = -\nabla V$ by off-path probing (§6.2) and form the subspace Jacobian $\partial_i F_j$. *Pass:* its antisymmetric part ≈ 0; F is a gradient, a scalar potential exists. *Fail:* nonzero stable curl; no scalar potential — the force has a rotational part.

Test 3 — reparameterization robustness. Recompute the candidate anchor score under a different time-clock (e.g. arc-length instead of token-count). *Pass:* the score's anchor ranking is stable. *Fail:* the score changes materially with the clock, i.e. it is an artifact of the (unphysical) token parameterization, often correlated with span length.

> Test independence is real and useful: a purely rotational force does no work, so Test 1 can *pass* (energy conserved) while Test 2 *fails* (nonzero curl). The curl estimate in Test 2 is the most numerically fragile quantity in the whole program: the antisymmetric part of a finite-difference Jacobian is exactly where estimation noise concentrates. "Stable curl" must mean stable under varying the probe step ϵ and the subspace; transient curl is noise, not physics.

5 Branch-conditional models

Each test outcome routes to a specific, named, textbook framework. The rule is Occam: *use the least general framework consistent with the results*, because each generalization buys descriptive power with extra free functions and thus with renewed unfalsifiability risk. In every live branch the detector has the same shape — *observed impulse minus the impulse the baseline model predicts*, with the baseline fit on coasting — and only the included force and the read-out differ.

5.1 Branch A — natural Lagrangian (all pass)

Define: V from (2); $E = \frac{1}{2}v^\top G v + V$. **Shape:** essentially nothing; baseline is $dE/dt = 0$, threshold from the coasting spread. **Momentum:** kinetic = canonical = Gv. **Anchor score:** the cheap scalar $|dE/dt|$ (needs only V, no gradient probe), or the full impulse residual $\|G\Delta v + \nabla V\|_{G^{-1}}$. **Work analogy:** fully intact — anchor work is genuine energy injection. **Validate:** coasting $|dE/dt|\approx 0$; anchor contrast vs. nulls; the collapse-to-V2 ablation (remove the V term — if discrimination survives, the potential was dead weight and this is not really V3).

5.2 Branch B — dissipative, structured (Test 1 structured fail)

Define: V as above plus a Rayleigh dissipation function $\mathcal{F} = \frac{1}{2}v^\top C v$. **Shape:** fit the drag C by regressing coasting dE/dt onto $-2\mathcal{F} = -v^\top C v$ (diagonal, PSD-constrained); the baseline becomes $dE/dt_{\text{exp}} = -2\mathcal{F}(v_t)$. **Momentum:** still $p = Gv$. **Anchor score:** the residual $dE/dt + 2\mathcal{F}$ — energy change beyond modeled drag. **Work analogy:** intact with richer bookkeeping; $\Delta T = W_{\text{cons}} + W_{\text{diss}} + W_{\text{anchor}}$, and you have modeled the first two. **Validate:** is the fitted C stable across traces? If not, linear drag is wrong and you may be heading to Branch E.

On the *dynamics* axis this is, in my honest estimate, the most likely landing: autoregressive generation has no obvious reason to be conservative, so expect Test 1 to structured-fail into here rather than into the clean Branch A. That is a claim about *which model fits the dynamics*, not about whether the detector then beats baselines — a dissipative system whose structure still does not predict anchors is a clean negative on the separate *outcome* axis (§9). Being in Branch B is the expected dynamics; beating baselines from it is the open question.

5.3 Branch C — gauge / rotational (Test 2 fail, Test 1 pass)

Define: Helmholtz-decompose the probed force in the active subspace, $F = -\nabla V_{\text{eff}} + F_{\text{sol}}$, and a vector potential via $F_{\text{sol}} = v \times B$, $B = \nabla \times A$. **Crucially redefine momentum** as the *canonical* momentum

$$p_{\text{can}} \;=\; Gv + A \;\neq\; Gv. \tag{4}$$

Shape: fit B (antisymmetric part of the subspace Jacobian, obtained from the same probes as ∇V) on coasting. **Anchor score:** $\|F_{\text{sol}}\|$, or the canonical-impulse residual; plus $|dE/dt|$ for any residual energetic anchors. **Work analogy: broken for the rotational anchors** — they do no work and are invisible to the work integral; you *must* use impulse. **Fingerprint:** bending without energy change — a momentum shift with no reinforcement. **Validate:** curl stable under ϵ and subspace; F_{sol} concentrates at labeled anchors.

This branch's central object, B, is the least trustworthy quantity in the plan. It can be written in five lines and still be pure finite-difference noise. If you land here, the first job is to establish that the curl is real, before believing any anchor result built on it.

5.4 Branch D — Maupertuis / Jacobi (Test 3 fail, Test 1 pass)

Define: fix the conserved energy level E (median of $T + V$ on coasting); the Jacobi metric $g_E = 2(E - V)\,G$; recast motion as a geodesic of g_E. **Shape:** estimate E and the metric field along the path. **Anchor score:** geodesic deviation in g_E, $\|\nabla_u^{g_E} u\|$ with u the arc-length-normalized velocity — clock-free by construction. **Work analogy: dissolved into geometry** — work and energy are absorbed into the conformal factor $2(E - V)$; momentum survives as the geodesic tangent. **Cost:** g_E is position-dependent, so Christoffel symbols return — but now they are the point, and they buy reparameterization invariance. **Validate:** does the score now survive the clock change that sent you here? *Dependency:* Maupertuis needs a conserved E, so this branch requires Test 1 to have passed; if Tests 1 and 3 both fail, Jacobi does not apply and the dissipative reparameterization-covariant generalization is required (much harder).

5.5 Branch E — metriplectic / GENERIC (multiple structured fails)

Define: a reversible (Poisson) bracket generated by E plus an irreversible bracket generated by an entropy S, with the degeneracy conditions $\{\cdot, S\} = 0$, $[\cdot, E] = 0$. **Shape:** both the conservative dynamics and the dissipative bracket — many free functions. **Anchor score:** anomaly in the reversible part against the modeled irreversible background. **Work analogy:** survives but fragments into reversible/irreversible parts.

The free-function count here is high enough that a positive result is hard to trust unless the simpler branches were cleanly ruled out *and* the GENERIC structure is strongly constrained by data. This is a last resort, not a destination.

5.6 Branch F — unstructured collapse (Test 1 unstructured fail)

Define: nothing physical. $p = Gv$ becomes one feature among $\{p, \Delta p, v, \text{surprisal}\}$ fed to a plain supervised anchor predictor. **Anchor score:** the predictor's output. **Both analogies: dead. Validate:** must beat position + surprisal + backtrack-keyword; if not, drop the approach.

The honest discipline in Branch F is to *say* the physics is dead and not relabel a logistic regression as mechanics. No metric, energy, or geometry helps once Test 1 fails without structure.

5.7 Summary of where each analogy survives

Branch	Momentum	Work	Read-out
A natural Lagrangian	Gv	intact	energy injection
B dissipative	Gv	intact*	drag-corrected injection
C gauge / rotational	$Gv + A$	blind	solenoidal bending
D Maupertuis / Jacobi	geodesic tangent	dissolved	geodesic deviation
E GENERIC	Gv	fragmented	reversible-part anomaly
F unstructured	feature only	dead	supervised prediction

*intact after subtracting the modeled dissipative work baseline. Momentum (impulse) survives in five of six branches; literal momentum–work survives cleanly in only two (A, B).

6 Implementation: variable-level details

This section pins each variable to an exact model quantity and marks every parameter as **pinned** (read directly), **sweep** (empirical, no correct a-priori value), or **measure** (a diagnostic that determines a ceiling, not a knob to tune for performance).

6.1 Shared skeleton (all branches)

State h_t (pinned read; sweep layer/position)

`hidden_states[layer]` from a forward pass with `output_hidden_states=True`; the residual stream after block `layer`. Two sub-choices: *layer* ($\sim$0.7 depth default, **sweep**) and *position* — the token-window span (§1.6) is represented by its last token by default (or its first-content token for the early decision, §2.4); mean pooling over the span is an ablation.

Velocity v_t (defined; sweep window)

Savitzky–Golay first derivative over a symmetric window; the forward half is the lag budget. Parameters: half-width w (**sweep**), polynomial order (2–3). *Units caveat:* compute per-token with $dt = 1$ and integrate over the span; do not mix a token clock with span-level decisions silently.

Metric G (defined; sweep shrinkage)

Constant whitening metric $G = \Sigma^{-1}$, Σ the velocity covariance *pooled over a held-out trace corpus* (not the scored trace — avoids leakage), with Ledoit–Wolf shrinkage (coefficient **sweep**) before inversion, since Σ is rank-deficient in $d \sim 5000$.

Momentum $p_t = Gv_t$ (pinned)
Covector.

Coasting set (measure/define; see §6.6)
Operational label of baseline steps; required to fit every branch baseline.

6.2 The potential V and its gradient

$V(h_t)$ (pinned read)
Summed token surprisal within the step: $-\sum_{\text{tok}} \log P_\theta(\text{tok})$, where logprobs are `log_softmax(logits)` gathered at the realized tokens. As a field: splice h at position t, run the upper layers, read the realized token's logprob.

$\nabla V(h_t)$ (estimated; sweep ϵ)
No closed form. Central finite difference along each active-subspace basis direction e_i: $\partial_i V \approx [V(h_t + \epsilon e_i) - V(h_t - \epsilon e_i)]/2\epsilon$. Each probe runs only layers $L{+}1 \to$ unembedding, reusing the KV-cache for positions $< t$. Cost: $\approx k\,(1 - \text{layer\_frac})$ forward-pass equivalents per probed step; probe at span decision points only (§2.4). ϵ: **sweep** (too small $\to$ float noise; too large $\to$ leaves the linear regime).

> The off-path splice (overwrite the hidden state at layer L, run upward with the prefix KV-cache) is the one piece I can write *structurally* but cannot guarantee runs unmodified: the API for partial-forward-from-a-layer with cache reuse is model- and `transformers`-version-specific. This is engineering, not physics, but it is the load-bearing engineering seam.

6.3 The active subspace

Basis $\{e_i\}_{i=1}^{k}$ (defined; measure k)
Top-k principal directions of the pooled, held-out *velocity* series (alternative: top force-field directions). k is set by the *energy capture* diagnostic: the fraction of velocity (or force) energy in the top k; choose the knee and **report the captured fraction as the recall ceiling**. k also sets the ∇V probe cost, so cost and accuracy are the same knob.

6.4 Fisher / KL proxy (only if a branch needs a non-trivial metric)

KL-proxy step length (pinned read; approx in K)
Replace $v^\top G^{\text{Fisher}} v$ by the measured $2\,\mathrm{KL}(P_{t-1} \| P_t)$ between consecutive boundary distributions; exact as the retained top-K vocab slice $\to$ full vocab. K: **sweep**.

> A *constant* whitening metric is the right default and keeps the EL residual flat (no Christoffel term). Position-dependent Fisher reintroduces the $\frac{1}{2}(\partial_i G_{jk}) v^j v^k$ kinetic-connection term and the associated estimation of ∂G; adopt it only if a constant metric demonstrably leaves signal on the table. Earlier in the design this Fisher term was dropped by mistake in the residual — it is only required *because* of a position-dependent metric, which is itself optional.

6.5 Branch-specific objects

Drag C (B)
least squares of coasting dE/dt on $-v^\top Cv$, diagonal PSD.

B-field (C)
antisymmetric part of the subspace Jacobian $\partial_i F_j$ — obtained from the *same* probe set as ∇V (one probe set yields the full Jacobian; symmetric part informs ∇V, antisymmetric part is B).

Energy level E (D)
median of $T + V$ over coasting.

Brackets $\{,\}$, $[,]$ and entropy S (E)
fit; high free-function count.

6.6 The coasting set — circularity, handled

The baseline fit in every branch needs coasting labels, but a clean label may not exist a priori. Bootstrap: (i) provisional Euclidean-energy score; (ii) take the calm bottom half as coasting; (iii) fit the baseline; (iv) recompute and check stability under one iteration. Alternatives: the constructive on-path/off-path labels from the synthetic curriculum (§2.5; off-path spans are coasting by construction), or low-surprisal runs.

> This bootstrap is circular by construction. It can be made stable-in-practice but not principled. State it as a limitation; do not let a coasting set chosen by a provisional score silently define the anchors the final score then "discovers."

7 Tiered execution plan

Everything specified above is something to try *eventually*; it is not a single experiment. The program runs in **tiers**. Tier 1 is the most basic, bells-and-whistles-free run that has a real chance of working on intuition and what is already known — but it still carries every *non-negotiable safeguard*, because omitting one of those is itself a critical failure, not a simplification. Each later tier adds a *small* subset of features chosen by the previous tier's outcome: if a tier failed, add the specific *fix* that addresses the failure observed; if it succeeded, skip fixes and add only an *improvement*. Tiers accrete until either every necessary fix has been tried and all fail (a clean negative), or the best working version is reached with every applicable improvement folded in. Nothing here changes the experiments or design of §§2–9; it only sequences them.

7.1 Three buckets every feature falls into

- **Always-on safeguards** (in *every* tier; omitting one is a critical failure, not a tier choice): counterfactual-importance resampling labels, not proxies (§2.1, §1.5); a *frozen semantic-difference adjudicator* so only genuinely different resamples count toward A_i (§2.2) — the specific adjudicator may be swapped for a cheaper one at larger scale, but it is fixed per label set, never tuned; exact token alignment via stored generated ids (§2.3); a frozen span definition (§1.6); held-out covariance for the metric, no leakage (§6); the constructive sanity check run *first* (§2.5); the coasting-fit measurement and the causal-vs-full- information ceiling measured *before* tuning (§4, §9); and the published baselines reproduced on the same traces (§3).

- **Fix-features** (added only to address a specific observed failure): the branch models keyed to which validity test failed — Branch C for nonzero curl, Branch D for a clock artifact, Branch E for combined failures (§5); the off-path ∇V probe (§6.2), which a fix needs the moment Test 2/Branch C enters; the position-dependent Fisher metric (§6); and the $D \to D^{\mathrm{IG}} \to$exact-ablation ladder for a first-order gradient that under-reads. (Branch B, the dissipative case, is *not* a later fix: it is one of the two Tier-1 detector options, selected by Test 1 — see below.)
- **Improvement-features** (added only on success). Two kinds, kept distinct because they pull in opposite directions on the accuracy axis: *capability* improvements buy something the detector could not do before but are not expected to raise accuracy — chiefly the first-content-token early decision and its lead-time/accuracy *curve* (§2.4), which by the causal-ceiling argument *trades* accuracy for lead time; *accuracy* improvements aim to raise the score itself — the full impulse residual $\|G\Delta v + \nabla V\|_{G^{-1}}$ in place of the cheap scalar (§5, Branch A/B), active-subspace tuning toward the energy-capture ceiling, and the KL/Fisher metric upgrade where it demonstrably adds signal (§6).

7.2 Tier 1 — the minimal run

The cheapest configuration that still tests the core bet. Crucially, Tier 1's first job is *diagnostic*: Test 1 (coasting energy flatness) reads off whether the system is conservative or dissipative, and that reading — not a prior assumption — selects the Tier 1 detector.

- **Detector (chosen by Test 1, not assumed):** if coasting $dE/dt \approx 0$, Branch A with the cheap scalar $|dE/dt|$; if coasting dE/dt shows a structured (fittable) decay, Branch B with the drag-corrected residual $dE/dt + 2\mathcal{F}$. Both use only V and a constant whitening metric — *no* ∇V probe, no Fisher, no Christoffels — and the span-complete decision (not the early commit). Branch B adds exactly one fitted parameter (the drag C), and it is fitted *only* once Test 1 has confirmed there is dissipation to fit; fitting drag to a near-conservative system would invent structure.
- **Diagnostics:** Test 1 (energy flatness) and Test 3 (reparameterization robustness) — both cheap, no probe. Test 2 (curl) is *deferred*, because it requires the ∇V probe, a Tier 2+ feature.
- **Baselines, reproduced on the same traces and labels:** the length-corrected kinematic signal *and* the learned step-boundary classifier trained directly against counterfactual-importance labels (§3). The learned classifier is the stronger competitor and the one that decides whether the *task* contribution survives even if the physics does not; it must be in Tier 1, not deferred.
- **Everything in the safeguard bucket**, on.

This is the no-frills version: fewest moving parts, lowest cost, and a central quantity (dE/dt) that is essentially free to compute. Its honest expectation is Branch B: the document's estimate is that the system is dissipative (§5), so the drag-corrected residual — not the bare $|dE/dt|$ — is the better-targeted of the two Tier-1 detectors. That it is better-targeted is a claim about fitting the dynamics; whether it then beats the baselines is the separate open question (§9). Tier 1 is built so that arriving at the right detector costs one cheap diagnostic, not a wasted run.

7.3 The ladder

The previous tier's outcome selects the next tier's single increment.

Tier	Increment added	Entered when
1	Minimal detector, Branch A *or* B by Test 1 (drag-corrected residual if dissipative) + all safeguards + Tests $1, 3$ + both baselines	Always: start here.
2-fix	The fix keyed to the failed test: Branch D if Test 3 (clock) fails; the ∇V probe + Test 2 + Branch C if rotational structure is suspected (§5)	Tier 1 detector (A or B) does not beat baselines, or Test 3 fails.
2-imp	*Capability* improvement: first-content-token early decision + its lead-time/accuracy *curve* (§2.4) — adds online lead time, expected to *cost* accuracy, not raise it	Tier 1 detector beats baselines and Test 3 passes.
3-fix	Next fix on the stack: Fisher metric if constant whitening leaves signal; D^{IG}/exact ablation if the gradient under-reads; Branch E only if combined failures persist	Prior fix tier helped but is insufficient.
3-imp	*Accuracy* improvement: full impulse residual $\|G\Delta v + \nabla V\|_{G^{-1}}$ in place of the scalar; subspace/metric tuning toward the energy-capture ceiling	Prior tier worked and more accuracy is wanted.

The fix and improvement paths can interleave across tiers: a fix that turns a failure into a success hands off to the improvement path, and a working detector that later plateaus can take a targeted fix. The *branch selected by the validity tests* (§5) is exactly the fix-feature ladder, indexed by which test failed — the tiering and the test→branch logic are the same logic, framed as execution order.

7.4 Termination

Stop at the first of: (i) every necessary fix tried and all fail — a clean negative, reported as "the momentum framing is metaphor, not mechanism" (§9, Branch F); or (ii) the best working version — Tests pass, baselines beaten, and every applicable improvement folded in with no remaining increment that helps.

"Likely to succeed" is relative, and the document's own estimate shapes Tier 1: the most probable live outcome is the dissipative case (§5, Branch B), so Tier 1's expected landing is the drag-corrected residual, reached via one cheap diagnostic (Test 1) rather than a wasted Branch A run. Tier 1 passing on the bare conservative scalar would be the pleasant surprise, not the base case. The entire point of tiering is to *fail cheap* — to learn the core bet is wrong (or that the dissipative detector already suffices) before building the probe, the curl machinery, the early-decision curve, or any other bell or whistle. A tier that adds more than one feature at a time forfeits this: if two features go in together and the result moves, the cause is ambiguous. One increment per tier is the discipline that makes each outcome interpretable.

8 What can and cannot be implemented now

Fully implementable and testable here (pure NumPy, no model). The entire physics stage downstream of the extracted arrays: Savitzky–Golay velocity, whitening metric with shrinkage, momentum, impulse residual, energy and its rate, all branch baselines (drag fit, curl/B from the probe Jacobian, Jacobi geodesic deviation), the branch router, and all validation diagnostics (coasting flatness, curl stability, reparameterization robustness, collapse-to-V2 ablation). Synthetic-anchor tests confirm the arithmetic.

Writable correctly but not verifiable from here (needs the model/GPU). The extraction stage: teacher-forcing, boundary hidden-state reads, surprisal, and the ∇V probe. Structurally sound, but the partial-forward-from-layer splice with cache reuse is environment-specific and must be adapted.

Writable but numerically unvouchable until run on real traces. The curl/B computation (Branch C) and the gradient step ϵ: the code is correct; whether its output is signal or finite-difference noise is a property of the data, not the code.

"I can implement all the physics" is true at the level of *operations* and true-and-tested for the NumPy stage. It is *not* a claim that the curl will be meaningful, that the probe will run unmodified on your model, or — most importantly — that the idea works. The code is instrumentation; the first real result is the coasting-fit number.

9 What the results would mean

Test 1 passes, contrast beats baselines, ∇V ablation hurts. The strongest positive: reasoning is approximately a conservative least-action flow, anchors are energy-injection events, and the potential carries independent signal. "Beats baselines" here means beats the two concrete, published competitors named in §3 — a learned step-boundary classifier trained directly against counterfactual-importance labels (the atheoretical version of the *task*), and a length-corrected kinematic signal in the spirit of Gjølbye et al. — evaluated against counterfactual-importance labels, not proxies. If it clears those, it is a solid, publishable result: the first online anchor detector, with a physics instantiation that earns its keep. It is *not* field-changing; calibrate expectations accordingly.

Two distinct collapse tests, two distinct verdicts. The ∇V ablation and the kinematic-baseline comparison probe different failures. If removing ∇V does *not* hurt, V3 has *collapsed to V2*: the signal is in the momentum-change term alone, the potential is dead weight, and it should be reported as an impulse detector, not a least-action one. If the full V3 score does not beat the *length-corrected kinematic* baseline, it has *collapsed to geometry*: the mechanical formalism adds nothing over directness/curvature, and the honest report is "trajectory geometry predicts anchors," crediting the substrate literature. Either collapse still leaves the *task* contribution intact if the score beats chance and the proxies are excluded — but it removes the physics contribution.

Structured dissipative / rotational / clock-artifact (B/C/D). The mechanical analogy holds in a more general form; the corresponding branch is the honest model. A rotational (C) result is the most scientifically interesting *if* the curl is real: it would say anchors are re-directional, not energetic.

Unstructured fail (F). The conservative-system premise is wrong and no fittable dissipative structure replaces it: dE/dt is large and patternless, so no energy concept survives. This is *not* the expected dynamics (the dissipative Branch B is); it is the worst dynamics case, in which the physics is dead from the start.

Contrast within noise of baselines (the likely outcome). Independently of which branch the dynamics land in, the most likely *outcome* is that the detector does not beat the baselines:

anchors may have no local dynamical signature beyond what trivial features capture. This is a genuine, valuable negative — it tells you the momentum framing is metaphor, not mechanism, before any expensive scaling. Note it kills the *method*, not necessarily the *task*: a learned classifier might still predict anchor-influence online even where the physics does not.

The ceiling number. Independently, restrict any *full-information* score to causal information and measure how much anchor-predictive power survives. The gap is the price of streaming and caps the inference goal regardless of cleverness. Measure it early.

Bottom line. The construction is physics-informed in *structure* (mass=metric, momentum=covector, conservative force, energy integral — all textbook) but not physics-*derived* from the model's actual update rule. Three assumptions — the conservative premise, the token-clock, and surprisal-as-potential — are imposed, each with a concrete test that can fail. The most probable *outcome* is a clean negative (the detector not beating baselines), even though the most probable *dynamics* is the dissipative Branch B. The design's worth is that it reaches whichever verdict it reaches cheaply, with the failure modes named in advance rather than discovered after a large-model run.

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